

Image Steganography with CNN Based Encoder-Decoder Model

ABSTRACT

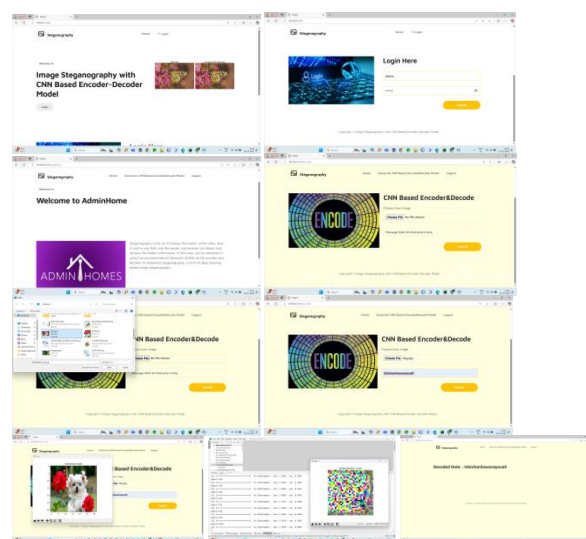
Image steganography research focuses on detecting hidden information within digital images. This paper proposes a novel method that combines a customized Convolutional Neural Network (CNN) with an encoder-decoder architecture for accurate and efficient steganalysis. The CNN extracts discriminative features from steganographic images, while the encoder-decoder network reconstructs the original image to identify embedding artifacts. By comparing the reconstructed image with the original using similarity measures, the model effectively detects hidden content. This approach enhances robustness and efficiency in digital forensics and security applications.

DESCRIPTION

This project uses a CNN-based Encoder-Decoder model to securely hide secret information inside digital images. The encoder embeds the message into the cover image, and the decoder accurately retrieves it without affecting image quality. The system improves security, robustness, and efficiency compared to traditional steganography methods.

1. Uses deep learning (CNN) for intelligent data embedding and extraction.
2. Maintains high image quality with strong resistance to attacks.
3. Provides high payload capacity and accurate message retrieval.
4. Adapts to diverse image datasets and dynamic communication needs.

Output:



EXISTING SOLUTIONS

- Uses traditional methods like LSB and DCT for hiding data.
- Has limited payload capacity and reduced robustness.
- Vulnerable to compression, noise, and other attacks.
- May affect image quality and lacks adaptability.

CONCLUSION

The CNN-based encoder-decoder model for image steganography offers a robust and efficient solution for secure data hiding. It outperforms traditional methods in terms of imperceptibility, payload capacity, and resistance to distortions. This system demonstrates the potential of deep learning in enhancing steganographic techniques.

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